Harry Zhu

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Education

University of Waterloo — Waterloo, ON
Honours Bachelor of Computer Science (Data Science) — *Apr 2028*

- Courses: Machine learning, Distributed Computing, Algorithms, Compilers, Linear Regression, Statistics

Technical Skills

Languages: Java, Rust, Scala, C++, C#, TypeScript, Python, SQL, R
Technologies: Spark, Delta Lake, DynamoDB, Kafka, Pandas, NumPy, TensorFlow/Keras, PyTorch, SciKit
Developer Tools: AWS, MySQL, SQL Server, PostgreSQL, Redis, Docker, Azure, Git

Experience

Amazon Prime Air — Sept 2026 – Dec 2026
Software Development Engineer — *Seattle, WA*

- Built an **LLM agent pipeline** that turns flight requirements into verification code run on drone flight logs, scaling translation to **2000+ safety requirements**.
- Collaborated with the **Stanford and NASA researchers behind ARTEMIS (ICSE 2026)** on a modified approach aimed at removing the human from the loop, using a test case oracle to completely eliminate ambiguities.
- Extended the LLM pipeline to generate **C++ runtime monitors** compiled into the drone's flight software and validated them in hardware-in-the-loop simulations.

Databricks — May 2026 – Aug 2026
Software Engineer — *Mountain View, CA*

- Developed **Databricks Real-time mode support for Delta Lake**, enabling low-latency Spark Structured Streaming over transactional Delta tables and expanding Real-time mode beyond traditional message queue sources.
- Reduced Delta streaming latency by **1640ms (66%) at P50** and **2790ms (41%) at P99** over traditional microbatch mode.
- **Decreased compute cost by up to 32%** for fixed-throughput workloads compared to microbatch mode, and reduced ZeroBus ingest latency allowing Delta Lake to replace Kafka workloads at roughly 50% of the cost.
- Architected a **high-performance distributed load balancing and scheduling algorithm** that partition processing work efficiently across Spark clusters and reduce load skew between workers.

Amazon Web Services — May 2025 – Dec 2025
Software Development Engineer — *Seattle, WA*

- Designed an auto re-enablement system for a **low-latency routing mechanism** in DynamoDB's routing layer, **increasing coverage from 73.5% to 90.8%** across all regions, enabling more customers to experience reduced latencies.
- Extended the mechanism to more data plane APIs, delivering a **750μs (~38.3%) reduction at P50** and **1870μs (~15.8%) reduction at P99** across all regions.
- Developed a **dynamic routing algorithm** that computes a moving average of observed network latencies to automatically route traffic to the fastest target, ensuring consistent low-latency performance and high availability.
- Redesigned a **reverse bias back-off algorithm** that detects overloaded targets and **intelligently redistributes traffic**, supporting the increase in Availability Zones and maintaining stability under high load.

Projects

ColdIVF | *Rust, Python, NEON, AVX*

- Built the **fastest vector index** for cold queries, beating faiss, DiskANN, LanceDB, Qdrant, usearch, AsterVec and SymphonyQG by **at least 2.6x in read speed and 3.4x less memory**.
- Engineered a single-file IVF index in Rust with dual quantization (1-bit RaBitQ and 4-bit codes), batched async disk reads, SIMD scans, durable writes, and Python bindings.

SpecWave | *Rust, Candle, Metal, Python*

- Built a **speculative search layer in Rust** on ColdIVF's fast batched cold reads, predicting the final embedding from a middle transformer layer and issuing those reads while the model still runs.
- Benchmarked on 9 embedding models including BERT, NomicBERT, Qwen3, and arctic, **cutting search time after the embedding by 92% at P50**, with no recall loss.